LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 1 Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations · Week 1 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each. Do Day 1 on the day you study the chapter, then Day 2 and Day 3 on the next two days.
2. Write your answers in your notebook, then check them with the key the same day. Copy every mistake into your error log.
3. Every equation you write must be balanced. Count every atom on both sides!

DAY 1: WRITING AND BALANCING EQUATIONS

Q1. Magnesium ribbon burns with a dazzling white flame and turns into a white powder. The white powder is:

(a) magnesium carbonate (b) magnesium hydroxide
(c) magnesium nitrate (d) magnesium oxide

Q2. We balance a chemical equation so that it obeys the law of:

(a) conservation of energy (b) constant proportions
(c) conservation of mass (d) multiple proportions

Q3. Which state symbol shows that a substance is dissolved in water?

(a) (s) (b) (l)
(c) (aq) (d) (g)

Q4. Heating limestone ($CaCO_3$) to make quicklime is an example of:

(a) a combination reaction
(b) a thermal decomposition reaction
(c) a displacement reaction
(d) a photolytic decomposition reaction

Q5. What does the arrow (→) in a chemical equation mean? On which side do we write the reactants?

Q6. Balance these equations:

(i) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
(ii) $Na + H_2O \rightarrow NaOH + H_2$
(iii) $Al + O_2 \rightarrow Al_2O_3$
(iv) $KClO_3 \rightarrow KCl + O_2$

Q7. Write a balanced equation with state symbols: solid zinc reacts with aqueous copper sulphate to give aqueous zinc sulphate and solid copper.

Q8. Hydrogen gas combines with chlorine gas to form hydrogen chloride gas. Write the balanced equation and name the type of reaction.

DAY 2: TYPES OF REACTIONS

Q9. Which of the following is a double displacement reaction?

(a) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ (b) $CaO + H_2O \rightarrow Ca(OH)_2$
(c) $2KClO_3 \rightarrow 2KCl + 3O_2$ (d) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

Q10. An iron nail is kept in copper sulphate solution for about 20 minutes. The nail:

(a) gets a brownish coating (b) becomes shiny and silvery
(c) dissolves completely (d) turns bright green

Q11. In the reaction $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, the substance that is reduced is:

(a) MnO_2 (b) HCl
(c) $MnCl_2$ (d) Cl_2

Q12. Why is respiration called an exothermic reaction? Write the balanced equation for the breakdown of glucose in our cells.

Q13. Silver bromide is used in black-and-white photography. Write the balanced equation for what happens when it is exposed to light, and name the type of reaction.

Q14. Name the type of each reaction and give a reason:

(i) $NaOH + HCl \rightarrow NaCl + H_2O$
(ii) $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$ (on heating)
(iii) $Mg + 2HCl \rightarrow MgCl_2 + H_2$

Q15. Zinc reacts with dilute hydrochloric acid, but copper does not. Write the balanced equation for the reaction that happens, and explain what this tells you about the reactivity of zinc and copper compared with hydrogen.

DAY 3: MIXED PRACTICE AND CHALLENGE

Q16. **Assertion (A):** Most decomposition reactions are endothermic.
Reason (R): Energy, as heat, light or electricity, is needed to break a compound into simpler substances.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q17. Which of these changes involve oxidation?

(i) rusting of iron (ii) rancidity of butter (iii) respiration (iv) melting of wax

(a) (i) and (ii) only (b) (i), (ii) and (iii)
(c) (ii) and (iv) (d) all four

Q18. A student dipped a clean copper wire into colourless silver nitrate solution. After an hour, the solution had turned blue, and a shiny greyish-white deposit had formed on the wire.

(i) Name the deposit on the wire. Why did the solution turn blue?

(ii) Write the balanced equation with state symbols.

(iii) Name the type of reaction. What does it tell you about copper and silver?

Q19. (a) Burning magnesium keeps burning in carbon dioxide: $2Mg + CO_2 \rightarrow 2MgO + C$. Which substance is oxidised and which is reduced?
(b) Why is hydrogen peroxide stored in dark-coloured bottles? Write the balanced equation.
(c) Write the balanced equation for the burning of methane (natural gas) and say whether it is exothermic or endothermic.

Q20. A student adds barium chloride solution to sodium sulphate solution in a test tube.

(i) What does she see?
(ii) Write the balanced equation with state symbols.
(iii) Name the type of reaction, and give one more example of the same type.

~ Three days, one chapter done. Now put your mistakes in the error log! ~